

Question 1: choose the correct answer:

- Which of the following refers to the steps of the knowledge discovery process, in which the several data sources are combined?
a. Data selection b. Data cleaning c. Data transformation d. Data integration
- Suppose one wants to predict the number of newborns according to the size of storks' population by performing supervised learning
a. Association rule b. Clustering c. Regression d. Classification
- Credit card fraud detection is an example of
a. Association rule b. Clustering c. Regression d. Classification
- An Unsupervised learning in which we Group data to form new categories
a. Association rule b. Clustering c. Regression d. Classification
- The following table is an example of
A. Transactional data b. data matrix c. database
d. Document data
- Attributes with the value {freshman, sophomore, junior, senior})
a. Ordinal b. Continuous C. Binary d. Nominal
- conditional probability** that a transaction having $\{X \cup Y\}$ also contains Z
a. support b. **Confidence** c. Count d. frequent itemset
- Given 4 items in transactional dataset, the possible candidate itemset
a. 32 b. 8 c. 16 d. 4
- The process of Handle missing data, smooth noisy data, identify or remove outliers
a. Data selection b. Data cleaning c. Data transformation e. Data integration
- predict the occurrence of an item based on the occurrences of other items in the transaction
a. Association rule b. Outlier Analysis. c. Regression d. Classification
- army rankings attribute type is
a. Ordinal b. Continuous C. Binary d. Nominal
- A measure for data quality in which some modified but some not
a. Consistency b. Interpretability c. Completeness d. Timeliness
- A measure for data quality in which how easily the data can be understood
a. Consistency b. Interpretability c. Completeness d. Timeliness
- Salary = "-10" is an example ofdata
a. Noisy b. incomplete c. Inconsistent
- A measure used to calculate true positive recognition rate
a. Recall b. precision c. f-measure d. Accuracy
- according to the following table to calculate information gain of outlook attribute
aa.0 b. 0.940 c. 0.971 d. 0.694

TID	Items
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

According to the following confusion matrix

Actual Class \ Predicted class	cancer = yes	cancer = no	Total
cancer = yes	TP 90	FN 210	300
cancer = no	FP 140	TN 9560	9700
Total	230	9770	10000

outlook	yes	no
rainy	2	3
overcast	4	0
sunny	3	2

- Precision value is
a. 0.39 b. 0.30 c. 0.34 d. 0.96
- Recall value is
a. 0.39 b. 0.30 c. 0.34 d. 0.96

For the following transactional database :

19. The Confidence of $\{I1, I2\} \Rightarrow \{I3\}$ is

- a. 0.66 b. 3 c. 0.75 d. 0.5

20. The Confidence of $\{I1, I3\} \Rightarrow \{I2\}$

- a. 6 b. 3 c. 0.5 d. 1

Transaction	List of items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

Q2: True or false

1. In association rule Implication means causality
2. In An association rule $\{Beer, Bread\} \rightarrow \{Milk\}$, $\{Milk\}$ is the antecedent of the rule
3. the main principle of Apriori algorithm is that: If an itemset is frequent, then all of its subsets must also be in frequent ✓
4. if $\{AB\}$ is a frequent itemset, both $\{A\}$ and $\{B\}$ should be a frequent itemset ✓
5. In Asymmetric binary attributes both outcomes are equally important ✗
6. In supervised learning the class labels of training data are unknown ✗
7. In classification using decision tree we select the attribute with the lowest information gain ✗
8. Temperature in C° or F° is an example of interval Attribute Types ✓
9. A numeric attribute is quantitative ✓
10. A ratio-scaled attribute is a numeric attribute with an inherent zero-point ✓

Q3: ANSWER THE FOLLOWING QUESTIONS:

1. Data in the real world is dirty. Discuss this statement
2. What is the difference between symmetric and Asymmetric binary data types
3. Define the main Apriori principle and the two main steps this Algorithm
4. List and Draw the Knowledge Discovery (KDD) Process steps.
5. List three data types with example